RISMS - Unit Test Report

71 of 73 unit tests pass (97.3%). The 2 failures reproduce known defects DEF-01 and DEF-02 from the System Test Report; no new defects were found at unit level.

1. Approach

Unit tests target the logic layer in isolation: `Validator`, `ProductRepository`, `ProductService`, `StockService`, `LedgerRepository`, `ReportService`, `AuthService`, `UserRepository`, `FileStore` and `ConfigStore`.

The suite is a single file, `tests/test_risms.cpp`.

2. Results by Unit

Unit under test	Tests	Pass	Fail
Validator	11	11	0
ProductService / ProductRepository (CRUD)	11	11	0
ProductService (search, find, list)	8	8	0
StockService	13	13	0
ReportService (history)	4	4	0
ReportService (inventory summary)	5	4	1
Persistence: ProductRepository, LedgerRepository, FileStore	9	8	1
AuthService / UserRepository	10	10	0
ConfigStore	2	2	0
Total	**73**	**71**	**2**

3. Test Log

UT ID	Test function	Unit	Req	Checks	Result
UT-01	`TC_PRD_01_validator_accepts_valid_product`	Validator	VFR02	Fully valid product returns Ok	Pass
UT-02	`TC_PRD_02_validator_accepts_blank_category`	Validator	BR02	Empty category is allowed	Pass
UT-03	`TC_PRD_04_validator_rejects_empty_sku`	Validator	VFR02	Empty SKU → `InvalidInput`	Pass

UT ID	Test function	Unit	Req	Checks	Result
UT-04	`TC_PRD_05_validator_rejects_empty_name`	Validator	VFR02	Empty name → `InvalidInput`	Pass
UT-05	`TC_PRD_06_validator_rejects_negative_price`	Validator	BR03	Price −5 rejected with exact message	Pass
UT-06	`TC_PRD_06b_validator_accepts_zero_price_boundary`	Validator	BR03	Price 0.00 accepted (boundary)	Pass
UT-07	`TC_PRD_09_validator_rejects_negative_threshold`	Validator	BR03	Threshold −2 rejected	Pass
UT-08	`TC_PRD_09b_validator_rejects_negative_quantity`	Validator	BR03	Quantity −1 rejected	Pass
UT-09	`TC_PRD_10_validator_rejects_pipe_in_name`	Validator	VFR02	`|` in name rejected	Pass
UT-10	`TC_PRD_10b_validator_rejects_pipe_in_sku_and_category`	Validator	VFR02	`|` in SKU and category rejected	Pass
UT-11	`TC_PRD_10c_validator_length_limits`	Validator	VFR02	Name 100 ok / 101 rejected; SKU 50 ok / 51 rejected	Pass
UT-12	`TC_PRD_01_add_assigns_sequential_ids`	ProductService	FR02	IDs P0001, P0002 assigned in order	Pass
UT-13	`TC_PRD_03_duplicate_sku_rejected`	ProductRepository	VFR01, BR01	Duplicate SKU → `DuplicateSku`, count unchanged	Pass
UT-14	`TC_PRD_03b_invalid_product_not_stored`	ProductService	VFR02	Invalid product never reaches repository	Pass
UT-15	`TC_PRD_11_update_changes_fields`	ProductService	FR03	Name and price updated, SKU kept	Pass
UT-16	`TC_PRD_12_update_to_existing_sku_rejected`	ProductRepository	BR01	Taking another product's SKU → `DuplicateSku`, original kept	Pass
UT-17	`TC_PRD_12b_update_keeping_own_sku_allowed`	ProductRepository	FR03	Re-saving own SKU is not a duplicate	Pass
UT-18	`TC_PRD_13_update_unknown_product`	ProductService	FR03	Unknown ID → `ProductNotFound`	Pass
UT-19	`TC_PRD_13b_update_requires_id`	ProductService	FR03	Empty ID → `InvalidInput`	Pass
UT-20	`TC_PRD_14_remove_existing_product`	ProductService	FR04	Removed product no longer found	Pass
UT-21	`TC_PRD_17_remove_unknown_product`	ProductService	FR04	Unknown ID → `ProductNotFound`	Pass
UT-22	`TC_PRD_18_ids_not_reused_after_removing_middle`	ProductRepository	BR01	After removing P0002 of 3, next ID is P0004	Pass
UT-23	`TC_SRC_01_search_exact_name`	ProductService	VFR03	Exact name returns one match	Pass
UT-24	`TC_SRC_02_search_case_insensitive`	ProductService	FR05	`RICE` matches `Rice 5kg`	Pass
UT-25	`TC_SRC_03_search_partial_match_multiple`	ProductService	FR05	`rice` matches 2 of 3 products	Pass
UT-26	`TC_SRC_04_search_no_match`	ProductService	VFR04	Unmatched text returns empty	Pass
UT-27	`TC_SRC_05_find_by_id_returns_all_fields`	ProductService	FR06	All stored fields returned	Pass
UT-28	`TC_SRC_05b_find_by_unknown_id`	ProductService	VFR04	Unknown ID → `ProductNotFound`	Pass
UT-29	`TC_SRC_06_list_all`	ProductService	FR11	Lists both products	Pass
UT-30	`TC_SRC_07_list_empty`	ProductService	FR11	Empty inventory lists nothing	Pass
UT-31	`TC_STK_01_stock_in_increases_quantity`	StockService	VFR05, BR06	0 + 20 = 20, then + 5 = 25	Pass
UT-32	`TC_STK_02_stock_in_is_logged`	StockService	VFR07, BR08	Movement M000001 has ID, type, change, user, reason, timestamp; one ledger line	Pass
UT-33	`TC_STK_03_stock_in_rejects_zero_and_negative`	StockService	SEC-03	0 and −3 rejected, no ledger entry	Pass

UT ID	Test function	Unit	Req	Checks	Result
UT-34	`TC_STK_05_stock_in_unknown_product`	StockService	FR07	Unknown ID rejected, no ledger entry	Pass
UT-35	`TC_STK_05b_stock_in_overflow_rejected`	StockService	NFR03	Stock-in beyond `INT_MAX` rejected, qty unchanged	Pass
UT-36	`TC_STK_06_stock_out_decreases_quantity`	StockService	FR07	20 − 15 = 5, change logged as −15	Pass
UT-37	`TC_STK_07_stock_out_exceeding_stock_rejected`	StockService	VFR06, BR07	Out 50 of 20 → `InsufficientStock`, nothing logged	Pass
UT-38	`TC_STK_08_stock_out_exact_quantity_boundary`	StockService	BR07	Out 10 of 10 → qty 0	Pass
UT-39	`TC_STK_09_stock_out_from_zero_rejected`	StockService	BR07	Out 1 of 0 rejected	Pass
UT-40	`TC_STK_09b_stock_out_rejects_zero_and_negative`	StockService	SEC-03	0 and −1 rejected	Pass
UT-41	`TC_STK_10_check_stock`	StockService	FR08	Returns current quantity 7	Pass
UT-42	`TC_STK_11_check_stock_unknown_product`	StockService	FR08	Unknown ID → `ProductNotFound`	Pass
UT-43	`TC_STK_15_ledger_failure_rolls_back_quantity`	StockService	SEC-04, NFR03	Ledger write fails → qty rolled back in memory and on disk	Pass
UT-44	`TC_STK_12_history_newest_first`	ReportService	FR10	M000002 (OUT) listed before M000001 (IN)	Pass
UT-45	`TC_STK_12b_history_filters_by_product`	ReportService	FR10	Only the selected product's movements returned	Pass
UT-46	`TC_STK_13_history_empty`	ReportService	FR10	Product with no movements returns empty	Pass
UT-47	`TC_STK_14_history_unknown_product`	ReportService	FR10	Unknown ID → `ProductNotFound`	Pass
UT-48	`TC_SUM_01_summary_totals`	ReportService	FR12	1 product, 5 units, ₹2,250.00, 1 low-stock	Pass
UT-49	`TC_SUM_02_qty_equal_threshold_is_low`	ReportService, Product	VFR09, BR10	qty 5 = threshold 5 is low in both	Pass
UT-50	`TC_SUM_03_qty_above_threshold_not_low`	ReportService, Product	VFR09, BR10	qty 12 > threshold 5 is not low	Pass
UT-51	`TC_SUM_04_listing_and_summary_agree_for_zero_threshold`	ReportService, Product	VFR09, NFR03	Threshold 0, qty 5: listing and summary agree	**Fail** (DEF-01)
UT-52	`TC_SUM_05_summary_empty_inventory`	ReportService	FR12	All totals 0	Pass
UT-53	`TC_PER_01_product_written_to_file`	ProductRepository	FR13	File line is `P0001|SKU-001|Rice 5kg|Grocery|450.00|0|5`	Pass
UT-54	`TC_PER_03_data_reloaded_after_restart`	ProductRepository, LedgerRepository	FR14, VFR08	New instances reload qty 12, price, history	Pass
UT-55	`TC_PER_03b_movement_ids_continue_after_restart`	LedgerRepository	BR08	Next movement after reload is M000002	Pass
UT-56	`TC_PER_04_missing_files_load_empty`	ProductRepository, LedgerRepository	FR14	Missing files load as empty, no error	Pass
UT-57	`TC_ROB_01_corrupt_product_line_skipped`	ProductRepository	NFR03	2 malformed lines skipped, valid one kept	Pass
UT-58	`TC_ROB_02_corrupt_ledger_line_skipped`	LedgerRepository	SEC-04	Malformed line and bad type `SIDEWAYS` skipped	Pass
UT-59	`TC_ROB_03_pipe_in_reason_consistent_before_and_after_restart`	StockService, LedgerRepository	SEC-04	Reason with `|` reads the same before and after reload	**Fail** (DEF-02)
UT-60	`TC_ROB_04_atomic_write_leaves_no_temp_files`	FileStore	NFR04	No `.tmp` files left after saves	Pass
UT-61	`TC_ROB_05_filestore_roundtrip_and_append`	FileStore	FR13	Creates nested dirs; write 2 + append 1 = 3 lines	Pass

UT ID	Test function	Unit	Req	Checks	Result
UT-62	`TC_AUTH_02_valid_login`	AuthService	FR01, SEC-01	Correct credentials return the user	Pass
UT-63	`TC_AUTH_02b_password_not_stored_in_plaintext`	AuthService, UserRepository	SEC-01	`users.txt` does not contain the password	Pass
UT-64	`TC_AUTH_03_wrong_password_counts_attempts`	AuthService	SEC-01	Messages show 2 then 1 attempts remaining	Pass
UT-65	`TC_AUTH_04_lockout_after_max_attempts`	AuthService	SEC-01	3rd failure → `AccountLocked`	Pass
UT-66	`TC_AUTH_05_correct_password_rejected_while_locked`	AuthService	SEC-01	Correct password still locked out	Pass
UT-67	`TC_AUTH_05b_lockout_expires`	AuthService	SEC-01	With 0 s lockout, next correct login succeeds	Pass
UT-68	`TC_AUTH_05c_success_resets_failed_counter`	AuthService	SEC-01	2 failures + success resets count	Pass
UT-69	`TC_AUTH_06_unknown_user`	AuthService	SEC-01	Unknown user → `UserNotFound`	Pass
UT-70	`TC_AUTH_08_lockout_persists_across_restart`	AuthService, UserRepository	SEC-01, BR09	Reloaded repository keeps the lock	Pass
UT-71	`TC_AUTH_09_duplicate_user_rejected`	UserRepository	SEC-01	Second `admin` account rejected	Pass
UT-72	`TC_CFG_01_config_parses_values_comments_whitespace`	ConfigStore	2.6	Comments, spaces, malformed lines and bad ints handled; defaults used	Pass
UT-73	`TC_CFG_02_missing_config_file`	ConfigStore	2.6	Missing file → `FileIOError`	Pass

4. Failures

UT ID	Assertion output	Root cause	Defect
UT-51	`CHECK_EQ(listedLow, summaryLow) got [0] vs [1]`	`Product::isLowStock()` uses threshold 0; `ReportService::inventorySummary()` falls back to the default 10	DEF-01 (Medium)
UT-59	`got [bad/reason] vs [bad|reason]`	`LedgerRepository::serialize()` replaces `|` only in the saved line; the in-memory movement keeps `|`	DEF-02 (Low)

Both tests are kept in the suite as regression tests: they will pass once the fixes in the System Test Report are applied.